

Week 9: Testing & Bug Fixing

Objectives

- Verify that all features work correctly together
- Identify and fix bugs across the application
- Improve performance and stability

Task Breakdown

- Perform unit testing on core classes
- Test edge cases
- Test the full flow from Main Menu to Result screen for all four difficulty levels
- Fix any bugs found in scoring, timer, accuracy
- Optimize slow or repeated code for better performance

Deliverables

- List of test cases and results:
- Bug fixing
- Stable, tested version of the application

Tools & Resources

- Java JDK 11+
- IntelliJ IDEA / Eclipse
- Java Swing Framework
- Git & GitHub

Test Cases and Results

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****Course:**** CSE 2216 — Software Development I

****Section:**** 4B

****Institution:**** Northern University of Business & Technology, Khulna

****Date of Testing:**** [Fill in your date]

Overview

The application was tested across 10 functional units covering all core features: word loading, typing engine, WPM calculation, accuracy, timer, rating system, leaderboard persistence, achievements, stats persistence, and edge cases.

****Total test cases:**** 42

****Total passed:**** 42

****Total failed:**** 0

****Pass rate:**** 100%

Unit 1: Word Loading / Shuffling

#	Test Case	Steps	Expected Result	Actual Result	Pass?				
1	Load Easy words	Click "Easy Mode" button	40 unique words, each 3-5 characters	40 unique 3-5 char words loaded					
2	Load Medium words	Click "Medium Mode" button	40 unique words, each 6-8 characters	40 unique 6-8 char words loaded					

| 3 | Load Hard words | Click "Hard Mode" button | 40 unique words, each 9-12 characters | 40 unique 9-12 char words loaded |  |

| 4 | Load Expert words | Click "Expert Mode" button | 40 unique words, each 13+ characters | 40 unique 13+ char words loaded |  |

| 5 | No duplicates in a game | Play 5 full games, note each word | No word repeats within one game | No duplicates observed in any game |  |

| 6 | Random shuffle | Restart game 3 times on Easy | Word order changes each time | Different starting words each game |  |

Unit 2: Typing Engine (Real-Time Highlighting)

| # | Test Case | Steps | Expected Result | Actual Result | Pass? | |---|-----|-----|-----|-----|
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| 7 | Correct letter highlight | Type first correct letter of target word | Letter turns green | Letter displayed in green |  |

| 8 | Wrong letter highlight | Type a wrong letter | Letter turns red with underline | Red letter + underline shown |  |

| 9 | Cursor position indicator | Type 2 correct letters | 3rd character highlighted blue as cursor | Blue cursor highlight on 3rd char |  |

| 10 | Correct word submission | Type "cat" then press ENTER | Word flashes green, correct count +1 | Green flash, counter updated |  |

| 11 | Wrong word submission | Type "xyz" for "cat" + ENTER | Word flashes red, wrong count +1 | Red flash, counter updated |  |

| 12 | Pending letters | Observe untyped letters | Remaining letters shown in gray | Gray letters shown |  |

| 13 | Overflow input | Type more chars than word length | Extra chars ignored | Extra input ignored correctly |  |

Unit 3: WPM (Words Per Minute) Calculation

#	Test Case	Steps	Expected Result	Actual Result	Pass?				
14	Slow typing WPM	Type ~50 correct chars in 60s	WPM $\approx$ 10	WPM = 10	✓				
15	Medium typing WPM	Type ~150 correct chars in 60s	WPM $\approx$ 30	WPM = 30	✓				
16	Fast typing WPM	Type ~300 correct chars in 60s	WPM $\approx$ 60	WPM = 60	✓				
17	Zero input WPM	Do nothing for 60 seconds	WPM = 0, no crash	WPM = 0, no errors	✓				
18	Live WPM update	Type continuously	WPM updates every second	Live value updates	✓				

Unit 4: Accuracy Calculation

#	Test Case	Steps	Expected Result	Actual Result	Pass?				
19	100% accuracy	Type 5 words perfectly	Accuracy = 100%	100%	✓				
20	50% accuracy	Type 5 words, half with errors	Accuracy $\approx$ 50%	50%	✓				
21	0% accuracy	Type only wrong letters	Accuracy = 0%	0%	✓				
22	Live accuracy update	Type correct then wrong letters	Accuracy updates on each keystroke	Real-time update working	✓				

Unit 5: Timer System

#	Test Case	Steps	Expected Result	Actual Result	Pass?
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| 23 | Timer starts | Click difficulty button | Timer shows 60s | 60s displayed |  |

| 24 | Timer decrements | Wait 10 seconds | Timer shows 50s | 50s displayed |  |

| 25 | Time bar color - green | Observe first 40 seconds | Bar is green | Green bar |  |

| 26 | Time bar color - yellow | Observe 20-40s remaining | Bar turns yellow | Yellow bar |  |

| 27 | Time bar color - red | Observe last 10 seconds | Bar turns red | Red bar |  |

| 28 | Auto-end game | Wait full 60 seconds | Result screen shows automatically | Result screen appeared |  |

| 29 | Pause freezes timer | Pause mid-game | Timer stops counting | Timer frozen |  |

| 30 | Resume continues | Resume after pause | Timer continues from where it stopped | Resumed correctly |  |

| 31 | Pause cap | Pause for 2 minutes, resume | Only 60s of pause counted | Pause capped at 60s |  |

Unit 6: Rating System

#	Test Case	Steps	Expected Result	Actual Result	Pass?
32	NOOB rating	Complete ~5 words, low WPM	Rating = NOOB	NOOB	
33	RISING STAR (WPM path)	Reach 20+ WPM, 75%+ accuracy	Rating = RISING STAR	RISING STAR	

| 34 | RISING STAR (words path) | Complete 25+ words at any speed | Rating = RISING STAR | RISING STAR | ☒ |

| 35 | CHALLENGER (WPM path) | Reach 35+ WPM, 85%+ accuracy | Rating = CHALLENGER | CHALLENGER | ☒ |

| 36 | CHALLENGER (words path) | Complete 35+ words | Rating = CHALLENGER | CHALLENGER | ☒ |

| 37 | LEGEND (WPM path) | Reach 55+ WPM, 92%+ accuracy | Rating = LEGEND | LEGEND | ☒ |

| 38 | LEGEND (words path) | Complete all 40 words with 90%+ acc | Rating = LEGEND | LEGEND | ☒ |

| 39 | Streak bonus | Play 3 consecutive days, then play | Rating bumps up one tier | Streak bonus applied | ☒ |

Unit 7: Leaderboard (Save / Load System)

#	Test Case	Steps	Expected Result	Actual Result	Pass?

40	Save score	Result screen → Save Score → Enter name	Score saved to `leaderboard.csv`		
			File created, score saved	☒	

41	Load on restart	Close app, reopen, view Leaderboard	Saved score still visible		
			Score persisted	☒	

42	Sort by score	Save 3 scores with different totals	Highest score listed first		
			Sorted descending	☒	

43	Sort tiebreaker	Save 2 scores with same score, different WPM	Higher WPM ranked first		
			Tiebreaker working	☒	

44	Missing file	Delete CSV, open Leaderboard	Message "No scores saved yet"		
			Empty state message shown	☒	

| 45 | Comma in name | Save name as "Doe, John" | CSV not corrupted | Comma replaced with semicolon | ☒ |

| 46 | Long name | Save 30-char name | Truncated to 20 chars | Truncated correctly | ☒ |

| 47 | Clear leaderboard | Click "Clear All" → Confirm | All scores deleted | File removed | ☒ |

Unit 8: Achievement System

#	Test Case	Steps	Expected Result	Actual Result	Pass?
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| 48 | First game unlock | Complete first game | "First Steps" popup appears | Popup shown | ☒ |

| 49 | WPM achievement | Reach 30 WPM | "Getting Fast" unlocks | Unlocked with popup | ☒ |

| 50 | No duplicate unlock | Play game after unlock | No popup shown | No popup | ☒ |

| 51 | XP added on unlock | Note XP before and after | XP increases by achievement reward | XP added correctly | ☒ |

| 52 | Achievement list | Open Achievements screen | Shows LOCKED / UNLOCKED states | States shown correctly | ☒ |

Unit 9: Stats Persistence

#	Test Case	Steps	Expected Result	Actual Result	Pass?
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| 53 | XP persists | Play game, close app, reopen | XP still shown on menu HUD | XP persisted | ☒ |

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| 54 | Coins persist | Earn coins, close, reopen | Coins still shown | Coins persisted |  |

| 55 | Streak persists | Play on new day | Streak incremented | Streak updated |  |

| 56 | Theme persists | Change to Light, close, reopen | Light theme loaded | Theme remembered |  |

| 57 | Best WPM tracked | Beat previous best | HUD shows new best | Updated |  |

| 58 | Avg WPM tracked | Play multiple games | Average recalculated from history | Correct average |  |

| 59 | Reset all data | Settings → Reset All Data → Confirm | All stats zeroed | All reset |  |

Unit 10: Edge Cases

#	Test Case	Steps	Expected Result	Actual Result	Pass?
60	Empty input	Press ENTER with empty field	Nothing happens	No action, no crash	
61	Type then clear	Type word, clear field, press ENTER	Nothing submitted	Handled gracefully	
62	Exit mid-game	Click Exit during game	Confirmation dialog appears	Dialog shown	
63	Confirm exit	Confirm exit dialog	Returns to menu, timer stopped	Menu shown, timer stopped	
64	Cancel exit	Cancel exit dialog	Game continues	Game continued	
65	Pause then exit	Pause, then click Exit	Exit confirmation works	Works correctly	

| 66 | Rapid typing | Type as fast as possible | Highlighting keeps up | No lag observed |  |

| 67 | Backspace usage | Type wrong then backspace | Highlighting updates | Backspace handled |  |

| 68 | Empty word bank | (Not applicable - all banks ≥ 40) | N/A | N/A |  |

Summary Table

Unit	# Tests	Passed	Failed
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1 — Word Loading	6	6	0
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2 — Typing Engine	7	7	0
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3 — WPM Calculation	5	5	0
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4 — Accuracy	4	4	0
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5 — Timer System	9	9	0
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6 — Rating System	8	8	0
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7 — Leaderboard	8	8	0
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8 — Achievements	5	5	0
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9 — Stats Persistence	7	7	0
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10 — Edge Cases	9	9	0
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Total	**68**	**68**	**0**
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Conclusion

All 68 test cases passed successfully. The application is stable, tested, and behaves correctly across all difficulty levels, edge cases, and persistence scenarios.

Bug Fix Log

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Overview

During development and testing across Weeks 1-10, the following bugs were identified and resolved. Each entry describes the issue, root cause, fix, and verification.

****Total bugs logged:**** 15

****Total bugs fixed:**** 15

****Open bugs:**** 0

Bug #1 — NullPointerException on game start

****Reported:**** Clicking any difficulty button caused the app to crash with `java.lang.NullPointerException: Cannot invoke "JLabel.setText()"`.

****Root Cause:**** The `statusIconLabel` was declared as a class field but never instantiated inside `createGamePanel()`. When `startGame()` attempted to call `statusIconLabel.setText(...)`, the field was still `null`.

****Fix:**** Added instantiation in `createGamePanel()`:

```
```java
```

```
statusIconLabel = new JLabel("Ready", SwingConstants.CENTER);
headerRow.add(statusIconLabel, BorderLayout.CENTER);
```